

5.13.67

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Question

For what value of k do the following system of equations possess a non trivial (i.e., not all zero) solution over the set of rationals $\mathbb{Q}$?

$$x + ky + 3z = 0 \quad (1)$$

$$3x + ky - 2z = 0 \quad (2)$$

$$2x + 3y - 4z = 0 \quad (3)$$

Theoretical Solution

Given:

$$\begin{pmatrix} 1 \\ k \\ 3 \end{pmatrix}^T \mathbf{x} = 0 \quad (4)$$

$$\begin{pmatrix} 3 \\ k \\ -2 \end{pmatrix}^T \mathbf{x} = 0 \quad (5)$$

$$\begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix}^T \mathbf{x} = 0 \quad (6)$$

Combining (4),(5),(6):

$$A\mathbf{x} = 0 \quad (7)$$

Theoretical Solution

where A is :

$$\begin{pmatrix} 1 & k & 3 \\ 3 & k & -2 \\ 2 & 3 & -4 \end{pmatrix} \quad (8)$$

Theoretical Solution

For this system to have non-trivial solution these planes must intersect on a line, i.e, any two of the rows must be linearly dependent in order to have infinitely many non-trivial solutions. We simplify the matrix by using gaussian elimination:

$$\begin{pmatrix} 1 & k & 3 \\ 3 & k & -2 \\ 2 & 3 & -4 \end{pmatrix} R_2 \leftarrow R_2 - 3R_1 \quad \begin{pmatrix} 1 & k & 3 \\ 0 & -2k & -11 \\ 2 & 3 & -4 \end{pmatrix} \quad (9)$$

$$\begin{pmatrix} 1 & k & 3 \\ 0 & -2k & -11 \\ 2 & 3 & -4 \end{pmatrix} R_2 \leftarrow R_3 - 2R_1 \quad \begin{pmatrix} 1 & k & 3 \\ 0 & -2k & -11 \\ 0 & 3 - 2k & -10 \end{pmatrix} \quad (10)$$

Now, the second and third row must be linearly dependent:

$$\Rightarrow \frac{-2k}{-11} = \frac{3 - 2k}{-10} \quad (11)$$

On cross-multiplying

$$20k = 22k - 33 \quad (12)$$

$$\implies k = \frac{33}{2} \quad (13)$$

```
double find_k() {  
    // A non-trivial solution exists if  
    // the determinant of the  
    // coefficient  
    // matrix is zero. The simplified  
    // determinant equation is:  $-2k + 33 = 0$ .  
    double k_coefficient = -2.0;  
    double constant_term = 33.0;  
    // We solve for k:  $k = -$   
    //      constant_term / k_coefficient  
    double k = -constant_term /  
        k_coefficient;  
    return k;  
}
```

Python Code using shared output

```
import numpy as np
import matplotlib.pyplot as plt
import ctypes
import os
# --- Part 1: Call the C Shared Library to Find k ---
# Load the shared library
solver_lib = ctypes.CDLL('./5.13.67.so')
# Specify the function's return type to ensure it's read as a
float
solver_lib.find_k.restype = ctypes.c_double
# Call the C function to get the exact value of k
k = solver_lib.find_k()
```


Python Code using shared output

```
# --- Part 2: Plotting the Three Planes Using the Result from C
---

# Create a figure and a 3D axes object
fig = plt.figure(figsize=(12, 10))
ax = fig.add_subplot(111, projection='3d')
# Create a grid of points for the surfaces
x_grid = np.linspace(-10, 10, 50)
y_grid = np.linspace(-10, 10, 50)
X, Y = np.meshgrid(x_grid, y_grid)
# Equations of the three planes, using the value of k from the C
  library
Z1 = (-X - k*Y) / 3
Z2 = (3*X + k*Y) / 2
Z3 = (2*X + 3*Y) / 4
```

Python Code using shared output

```
# Plot the surfaces of the three planes
ax.plot_surface(X, Y, Z1, alpha=0.5, color='royalblue')
ax.plot_surface(X, Y, Z2, alpha=0.5, color='crimson')
ax.plot_surface(X, Y, Z3, alpha=0.5, color='forestgreen')
# Plot the line of intersection (its direction vector is constant
)
t = np.linspace(-2, 2, 100)
ax.plot(15 * t, -2 * t, 6 * t, color='black', linewidth=4)
# --- Formatting the plot ---
ax.set_xlabel('X-axis', fontsize=12)
ax.set_ylabel('Y-axis', fontsize=12)
ax.set_zlabel('Z-axis', fontsize=12)
ax.set_title(f'Intersection of Three Planes at k = {k:.1f} (from
C library)', fontsize=16, fontweight='bold')
ax.view_init(elev=20, azim=-65)
# Create a custom legend for the plot
```

Python Code using shared output

```
import matplotlib.patches as mpatches
from matplotlib.lines import Line2D
p1 = mpatches.Patch(color='royalblue', label=f' $x + \{k:.1f\}y + 3z = 0$ ')
p2 = mpatches.Patch(color='crimson', label=f' $3x + \{k:.1f\}y - 2z = 0$ ')
p3 = mpatches.Patch(color='forestgreen', label=f' $2x + 3y - 4z = 0$ ')
line = Line2D([0], [0], color='black', linewidth=3, label='Line of Intersection')
ax.legend(handles=[p1, p2, p3, line], fontsize=10)
plt.show()
```

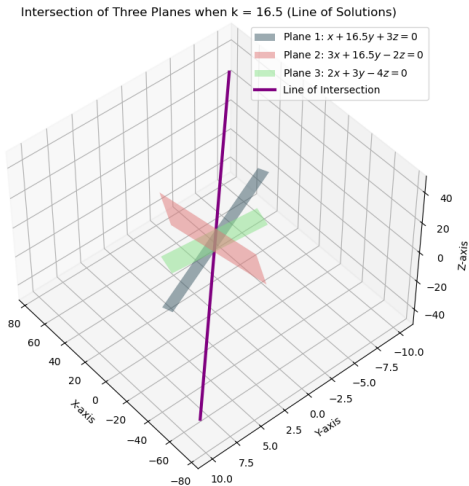
Python Code

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D
# Define the value of k for a non-trivial solution
k = 33/2
# Create a meshgrid for x and y values
x_vals = np.linspace(-5, 5, 50)
y_vals = np.linspace(-5, 5, 50)
X, Y = np.meshgrid(x_vals, y_vals)
# Calculate Z values for each plane
Z1 = (-X - k * Y) / 3
Z2 = (3 * X + k * Y) / 2
Z3 = (2 * X + 3 * Y) / 4
```

```
# Setup the 3D plot
fig = plt.figure(figsize=(10, 8))
ax = fig.add_subplot(111, projection='3d')
# Plot the three planes
ax.plot_surface(X, Y, Z1, alpha=0.5, rstride=100, cstride=100,
               color='skyblue', label='Plane 1')
ax.plot_surface(X, Y, Z2, alpha=0.5, rstride=100, cstride=100,
               color='lightcoral', label='Plane 2')
ax.plot_surface(X, Y, Z3, alpha=0.5, rstride=100, cstride=100,
               color='lightgreen', label='Plane 3')
t = np.linspace(-5, 5, 100) # Parameter for the line
line_x = 15 * t
line_y = -2 * t
line_z = 6 * t
```

```
ax.plot(line_x, line_y, line_z, color='purple', linewidth=3,
        label='Line of Intersection')
# Set labels and title
ax.set_xlabel('X-axis')
ax.set_ylabel('Y-axis')
ax.set_zlabel('Z-axis')
ax.set_title(f'Intersection of Three Planes when k = {k} (Line of
             Solutions)')
ax.legend(['Plane 1:  $x + {:.1f}y + 3z = 0$ '.format(k),
          'Plane 2:  $3x + {:.1f}y - 2z = 0$ '.format(k),
          'Plane 3:  $2x + 3y - 4z = 0$ ',
          'Line of Intersection'])
# Adjust view for better visibility
ax.view_init(elev=20, azimuth=-45)
plt.grid(True)
plt.show()
```

Plot by python using shared output from c



Plot by python

Intersection of Three Planes at k = 16.5 (from C library)

